U.S.N.										

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

December 2025 / January 2026 Semester End Main Examinations

Programme: B.E. **Semester: VI**
Branch: AI & DS/CSE(DS)/CSE(IOT) **Duration: 3 hrs.**
Course Code: 23DC6PCSEA **Max Marks: 100**
Course: Software Engineering & Agile Methodologies

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

		UNIT - I	*CO*	*PO*	**Marks**
1	a)	A compliance audit requires evidence that the software development lifecycle follows best practices. Classify the software metrics. Mention the real time use cases for each metrics.	*CO2*	*PO2*	**06**
	b)	Apply the RAD methodology to outline a development plan, with a neat diagram.	*CO1*	*PO1*	**08**
	c)	During the maintenance phase, it is discovered that the initial design does not support new regulatory requirements. What limitations of the traditional Waterfall model are exposed in this case, and how do extensions like the V-Model attempt to mitigate them?	*CO2*	*PO2*	**06**
		OR			
2	a)	A project manager observes that bugs are frequently found in the later stages of development and wants to improve the early detection of defects. Which software metrics could the manager introduce to monitor code quality and defect trends during early development phases? Justify your choice with a real-world implication.	*CO2*	*PO2*	**06**
	b)	With a neat diagram, elaborate on the Bohem's Spiral model. How does it incorporate risk analysis?	*CO1*	*PO1*	**08**
	c)	A company accustomed to using the Waterfall model is considering a switch to Agile for a new AI-based product.	*CO2*	*PO2*	**06**

			Compare the key differences between Agile and traditional models should the company consider, and how might these affect project planning and team dynamics?			
			UNIT - II			
	3	a)	A logistics software has separate modules for delivery tracking, customer notification, and route optimization. How would you design them to ensure high cohesion and low coupling?	*CO1*	*PO1*	**06**
		b)	Outline the functional and non-functional requirements for the following scenario, A ride-sharing app manages booking, driver assignment, fare calculation, and ride history.	*CO2*	*PO2*	**07**
		c)	A food delivery app receives orders, assigns riders, and processes payments. Create a DFD showing the interaction between users, restaurants, delivery staff, and payment systems.	*CO3*	*PO3*	**07**
			OR			
	4	a)	Imagine you are gathering requirements for a software project with unclear user expectations. Explain how prototyping can assist in requirements analysis and describe how you would mitigate potential limitations while using this technique.	*CO1*	*PO1*	**06**
		b)	Design a System Requirements Specification for the Library. The LMS will automate the process of borrowing, returning, and managing books, providing ease of access to library data for students, librarians, and administrators.	*CO2*	*PO2*	**07**
		c)	An airline launches a customer self-service portal for bookings, cancellations, and refunds. Create the context model with necessary external systems (e.g., payment processor, loyalty system).	*CO3*	*PO3*	**07**
			UNIT - III			
	5	a)	You are managing a project to build a mobile app for online grocery delivery. The app must be launched within 5 months. The stakeholders have insisted on the inclusion of real-time order tracking, which is a complex feature. How would you plan the project schedule to ensure timely delivery? Identify the essential steps for the project planning.	*CO2*	*PO2*	**06**
		b)	You are managing a software development project with the following tasks: 1. Requirement Gathering (3 days) 2. System Design (5 days, starts after requirements are gathered) 3. Module Development (10 days, starts after design) 4. Testing (6 days, starts after development)	*CO3*	*PO3*	**08**

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5. Bug Fixing (3 days, starts after testing)
6. Deployment (2 days, starts after bug fixing)

a) Create a Gantt chart to visualize the project timeline.
b) Build a PERT diagram and calculate the expected duration using optimistic (O), most likely (M), and pessimistic (P) estimates for each task.
c) Identify the critical path of the project.
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<tr><td></td><td></td><td>c)</td><td>
A software company is developing a project estimated at 20 KLOC. Estimate the effort, time, and team size by using COCOMO for all 3 development modes.
<table>
<tr><th>Software Projects</th><th>a</th><th>b</th><th>c</th><th>d</th></tr>
<tr><td>Organic</td><td></td><td>1.05</td><td>2.5</td><td>0.38</td></tr>
<tr><td>Semi-Detached</td><td>3.0</td><td>1.12</td><td>2.5</td><td>0.35</td></tr>
<tr><td>Embedded</td><td>3.6</td><td>1.20</td><td>2.5</td><td>0.32</td></tr>
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</td><td>CO3</td><td>PO3</td><td>06</td></tr>
<tr><td></td><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td></td><td>6</td><td>a)</td><td>Your team has previously completed 10 similar web application projects. How would you use empirical data from those projects to estimate effort and cost for a new web application of similar scope?</td><td>CO3</td><td>PO3</td><td>06</td></tr>
<tr><td></td><td></td><td>b)</td><td>An organization's average productivity is 12 FP/pm. The average labor rate is $15600 per month. If a proposed project has a count total of 560 and the Value Adjustment factor (VAF) is 34. Calculate and analyse

(I) Cost per Functional Point

(II) Overall Project Cost

(III) Estimated effort in person-months</td><td>CO2</td><td>PO2</td><td>08</td></tr>
<tr><td></td><td></td><td>c)</td><td>The company is starting a project involving data analytics, backend development, UI/UX design, and DevOps. The team is composed of specialists in each domain, but they’ve never worked together. Design the different team structure (functional, matrix, or cross-functional) and justify which would be most effective for this project.</td><td>CO3</td><td>PO3</td><td>06</td></tr>
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			UNIT - IV			
	7	a)	Discuss the significance of Agile values and principles in modern project management.	*CO1*	*PO1*	**06**
		b)	Interpret how a pull-based system works in Kanban. How does it differ from a push-based system in traditional project management? Provide suitable examples.	*CO2*	*PO2*	**06**
		c)	Your client *StyleSphere* wants to expand its online fashion store into the Middle East. The front-end will need localization, and logistics integration must align with regional partners. The client wants quick iterations and a fully functional MVP in 3 months using Agile Scrum. Create a scrum team from the following people. Justify your choice of ideal candidates. **Available Team Members:** ● Client Contact – Farah (insightful but limited time) ● Customer Representative – Zoya (based in region, responsive, understands fashion trends) ● Project Manager – Imran (strong in team coordination) ● Senior Developer – Ramesh ● Developer – Ankit (junior, UI specialist) ● Tester – Sofia (knows regional compliance rules) ● UX Designer – Iqbal (local design expertise) ● Scrum Master – Nisha (certified, well-liked)	*CO3*	*PO3*	**08**
			OR			
	8	a)	Interpret the various types of epics used in real-world JIRA implementations. Mention the use cases.	*CO2*	*PO2*	**06**
		b)	Identify one epic theme and convert the goals into appropriate user stories. In an online learning platform, the following goals are to be achieved: (i) Students should be able to enroll in available courses. (ii) Students should be able to watch recorded lectures and download resources. (iii) Instructors should be able to create and update course content.	*CO1*	*PO1*	**06**
		c)	You are building new features for an e-commerce website: ● Implement search bar functionality ● Design homepage banner ● Write test cases for cart functionality	*CO3*	*PO3*	**08**

			● Fix login bug reported by QA ● Review product listing code ● Deploy payment gateway integration to production a) Visualize a basic workflow (e.g., To Do → In Progress → Testing → Done). b) Categorize the items into each stage of the workflow. c) Create a Kanban board for this scenario and suggest WIP limits for each column			
			UNIT - V			
	9	a)	Outline the key phases of the Extreme Programming (XP) life cycle by analyzing how the XP life cycle supports rapid feedback and adaptability in software development.	*CO2*	*PO2*	**06**
		b)	Interpret the benefits and challenges of using a distributed Version Control System (e.g., Git) in a team of 50 developers working remotely. Suggest best practices to overcome the challenges.	*CO3*	*PO3*	**06**
		c)	Analyze a class diagram for an SCM system with classes like Configuration Item, Version, Change Request, and Repository. Evaluate its suitability for managing version control and change management.	*CO2*	*PO2*	**08**
			OR			
	10	a)	Evaluate the effectiveness of customer involvement in the XP life cycle by analyzing its influence on requirement gathering and project adaptability in dynamic environments	*CO2*	*PO2*	**06**
		b)	Your team is distributed across multiple time zones, and a change request is submitted to update a critical component of the system. The change requires input from developers, testers, and a database administrator, but coordinating across time zones is causing delays in the change approval process. Interpret the change management process to ensure timely review and approval of the change request while maintaining effective collaboration across the distributed team.	*CO3*	*PO3*	**06**
		c)	Analyze how a sequence diagram for an SCM system can model a distributed version control process across multiple repositories.	*CO2*	*PO2*	**08**

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